

Model Report: Multi-Channel CNN Patch-Level (5-seed ensemble)

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Generated: 2026-06-22 13:31:27

Script: /home/mmann1123/Documents/github/South_Africa_Crop_Comp/deep_learn/src/Deep Learning/Patch Level/Multi-Channel CNN Patch-Level (5-seed ensemble)

Data Split:

train_count: 21741
test_count: 830
random_seed: 42
split_method: field-based (patch-level)

Metrics (test set):

Accuracy: 0.7133
Cohen Kappa: 0.5974
F1 Weighted: 0.6961
F1 Macro: 0.6260

Training Time: 11288s (188.1 min)

Hyperparameters:

target_size: [128, 128]
batch_size: 64
epochs: 20
optimizer: Adam
loss: sparse_categorical_crossentropy
channels: 60
n_models: 5
seeds: [42, 101, 202, 303, 404]
aggregation: mean softmax over 5 seeds (patch-level), then majority vote by FID
per_seed_seconds: {'42': 2132.3, '101': 2498.5, '202': 2112.9, '303': 2154.9, '404': 2389.7}

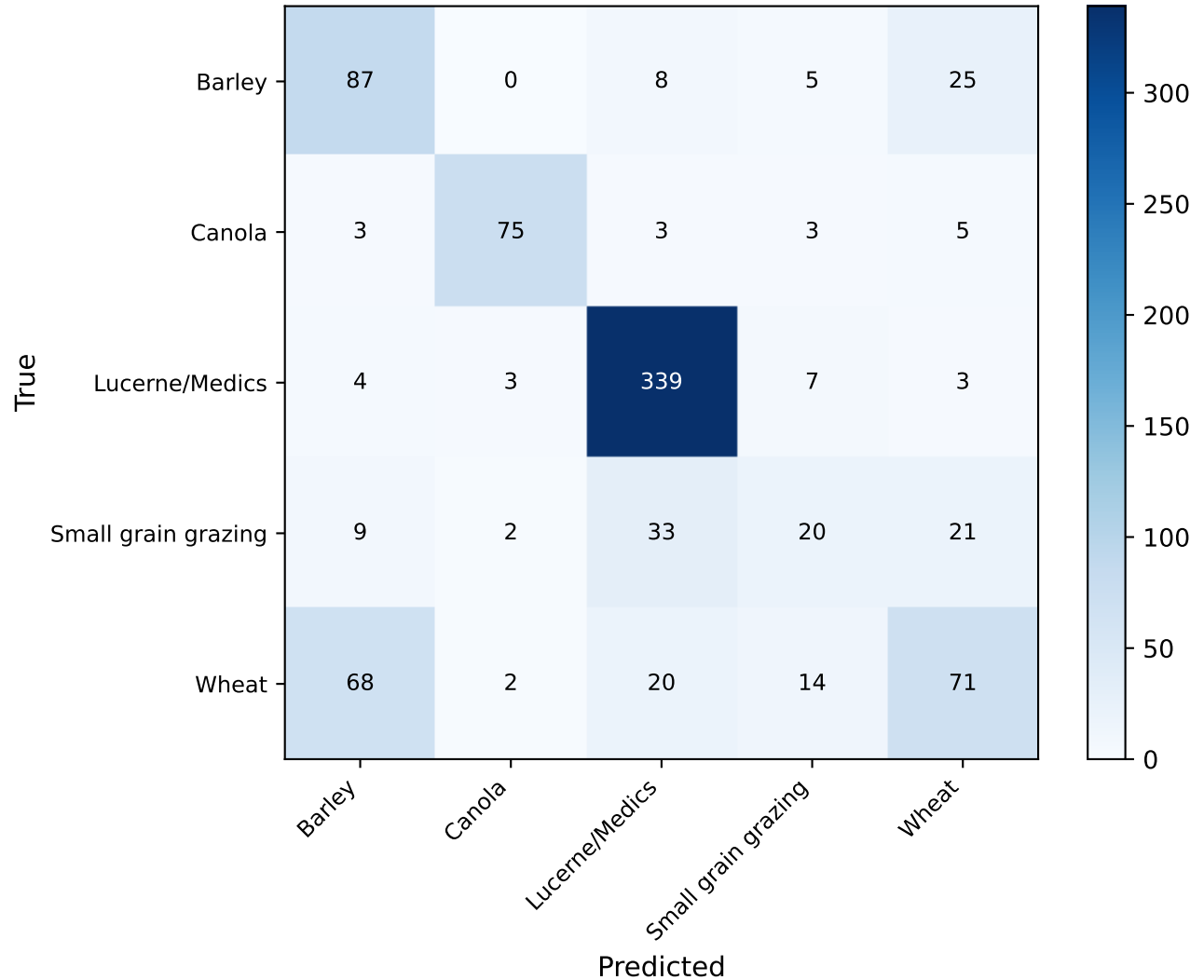
Notes:

Per-seed training time (s): {'42': 2132.3, '101': 2498.5, '202': 2112.9, '303': 2154.9, '404': 2389.7}

Per-Class Metrics

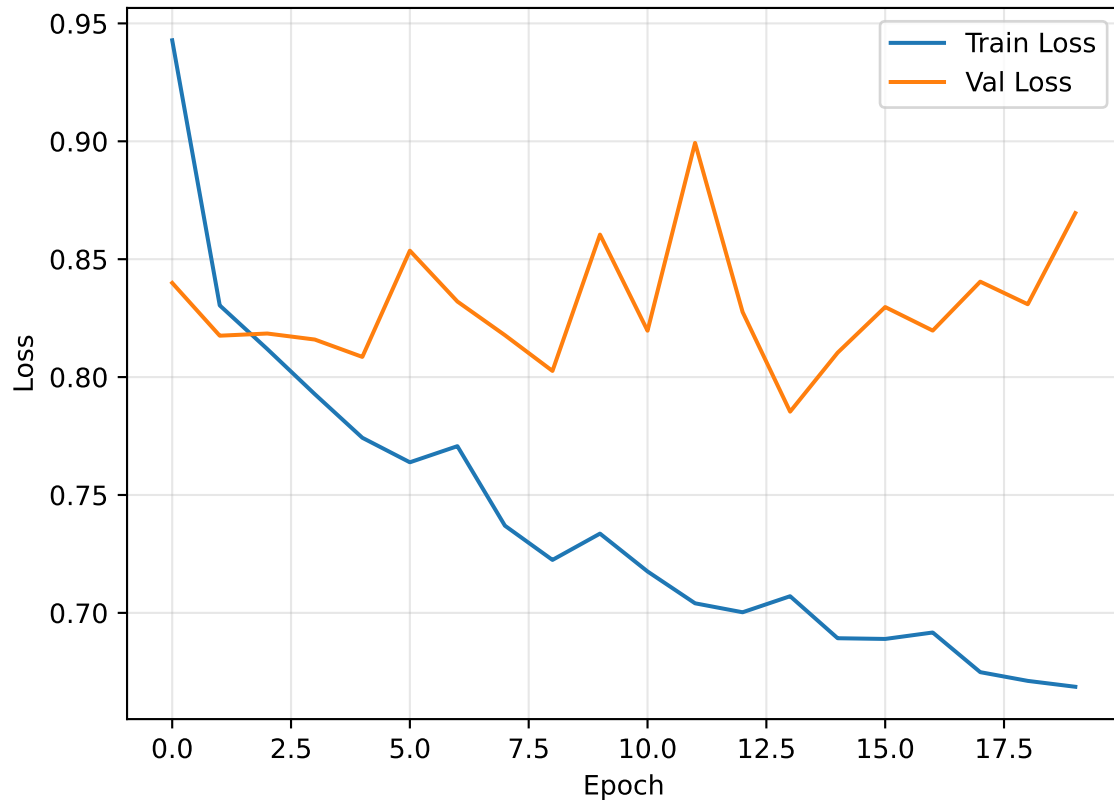
Class	Precision	Recall	F1-Score	Support
Barley	0.509	0.696	0.588	125
Canola	0.915	0.843	0.877	89
Lucerne/Medics	0.841	0.952	0.893	356
Small grain grazing	0.408	0.235	0.299	85
Wheat	0.568	0.406	0.473	175

Confusion Matrix



Training Curves

Training Loss



Training Accuracy

